

(Please write your Enrollment Number)

Enrollment No. \_\_\_\_\_

End-Term Examination  
(CBCS)(SUBJECTIVE TYPE)(OffLine)  
Course Name:<B.Tech>, Semester:<1>  
Reappear upto 2023 batch  
(December, 2024)

|                       |                              |
|-----------------------|------------------------------|
| Subject Code: BAS 109 | Subject: Applied Mathematics |
| Time :3 Hours         | Maximum Marks :60            |

Note: Q1 is compulsory. Attempt one question each from the Units I, II, III & IV.

| Q1     |                                                                                                                                                                                                                                                                                                                                                 | (2.5*8<br>=20) | CO<br>Mapping |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|---------------|
|        | a) If the Eigen values of $A = \begin{bmatrix} 4 & 0 \\ 5 & 7 \end{bmatrix}$ are 4 and 7 then find the eigen values of $A^2$ , $5A$ and $A^2 + 5A + I$                                                                                                                                                                                          |                | CO 1          |
|        | b) Find the rank of the matrix $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 4 & 1 & 6 \\ 3 & 6 & -3 & 9 \end{bmatrix}$ by elementary row transformation.                                                                                                                                                                                              |                | CO 1          |
|        | c) Find Laplace Transformation of $te^{2t} \cos 5t$                                                                                                                                                                                                                                                                                             |                | CO 2          |
|        | d) Find fourier coefficient $a_0$ , $b_n$ where $f(x) = x + x^2$ , $-\pi \leq x \leq \pi$                                                                                                                                                                                                                                                       |                | CO 2          |
|        | e) If $V = (x^2 + y^2 + z^2)^{-1/2}$ , then show that $x \frac{\partial V}{\partial x} + y \frac{\partial V}{\partial y} + z \frac{\partial V}{\partial z} = -V$ .                                                                                                                                                                              |                | CO 3          |
|        | f) Show that the function $f(x, y) = x^3 + y^3 - 63(x + y) + 12xy$ is maximum at $(-7, -7)$ and minimum at $(3, 3)$ .                                                                                                                                                                                                                           |                | CO 3          |
|        | g) Evaluate $\int_0^1 \int_1^2 \int_2^3 xyz \, dx \, dy \, dz$                                                                                                                                                                                                                                                                                  |                | CO 4          |
|        | h) Find grad where $\phi$ is given by $\phi = 3x^2y - y^3z^2$ at the point $(1, -2, -1)$ and Show that if $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ , $\text{div}(\vec{r}) = 3$ .                                                                                                                                                              |                | CO 4          |
| UNIT I |                                                                                                                                                                                                                                                                                                                                                 |                | CO<br>Mapping |
| Q2     | a). Using elementary row transformation, find the inverse of the matrix $\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 3 \\ 3 & 1 & 1 \end{bmatrix}$<br><br>b) Investigate the values of $\lambda$ and $\mu$ so that the equation $2x + 3y + 5z = 9, 7x + 3y - 2z = 8, 2x + 3y + \lambda z = \mu$ have (i) No solution (ii) A unique Sol (iii) Inf. sol. | (5*2=10)       | CO 1          |

|          |                                                                                                                                                                                                                                                                                                                                                          |          |            |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|
| Q3       | <p>a) Verify Cayley- Hamilton Theorem. &amp; also Find <math>A^{-1}</math> of <math>\begin{bmatrix} 1 &amp; 2 &amp; 0 \\ 2 &amp; -1 &amp; 0 \\ 0 &amp; 0 &amp; -1 \end{bmatrix}</math></p> <p>b) Find Eigen values and Eigenvectors of matrix <math>\begin{bmatrix} 3 &amp; 1 &amp; 4 \\ 0 &amp; 2 &amp; 6 \\ 0 &amp; 0 &amp; 5 \end{bmatrix}</math></p> | (5*2=10) | CO 1       |
| UNIT II  |                                                                                                                                                                                                                                                                                                                                                          |          | CO Mapping |
| Q4       | <p>a) Find the Inverse Laplace Transform of <math>F(s)</math> by convolution theorem where <math>F(s) = \frac{1}{(s^2+1)(s-2)}</math></p> <p>b) If <math>T:V_2(R) \rightarrow V_3(R)</math> defined by <math>T(a,b) = (a-b, a+b, 2b)</math> is a Linear Transformation then Find rank and nullity of <math>T</math></p>                                  | (5*2=10) | CO 2       |
| Q5       | <p>a) If <math>T: V_2 \rightarrow V_2</math> defined by <math>T(x, y) = (x+3y, x-4y)</math> is a Linear Transformation?</p> <p>b) Find Half range fourier cosine series of <math>f(x) = (x+\pi)/2, 0 \leq x \leq \pi</math></p>                                                                                                                          | (5*2=10) | CO 2       |
| UNIT III |                                                                                                                                                                                                                                                                                                                                                          |          | CO Mapping |
| Q6       | <p>a). If <math>u = \sin^{-1}\left(\frac{x}{y}\right) + \tan^{-1}\left(\frac{y}{x}\right)</math>, prove that <math>x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0</math></p> <p>b) State Leibnitz theorem and prove that <math>(1-x^2)y_2 - xy_1 - a^2y = 0</math>, If <math>y = e^{a \sin^{-1}x}</math></p>                      | (5*2=10) | CO 3       |
| Q7       | <p>a). Expand <math>x^2y + 3y - 2</math> in powers of <math>(x-1)</math> and <math>(y+2)</math> up to second degree terms.</p> <p>b). Find the extreme value of the function <math>f(x,y) = x^3 + y^3 - 3axy</math>. <math>a &gt; 0</math>.</p>                                                                                                          | (5*2=10) | CO 3       |
| UNIT IV  |                                                                                                                                                                                                                                                                                                                                                          |          | CO Mapping |
| Q8       | <p>a). Solve by changing the order of integration <math>\int_0^1 \int_{x^2}^{2-x} xy \, dx \, dy</math></p> <p>b). Find the directional derivative of the following function <math>f = 2x^2 + 3y^2 + z^2</math> at the point <math>P(2,1,3)</math> in the direction of the vector <math>\vec{a} = \hat{i} - 2\hat{k}</math>.</p>                         | (5*2=10) | CO 4       |
| Q9       | <p>a). Show that the vector field <math>\vec{f} = \frac{\vec{r}}{r^3}</math> is irrotational as well as solenoidal.</p> <p>b). Evaluate <math>\iiint_R (x-2y+z) \, dx \, dy \, dz</math>, where <math>R: 0 \leq x \leq 1, 0 \leq y \leq x^2, 0 \leq z \leq (x+y)</math>.</p>                                                                             | (5*2=10) | CO 4       |